

VATREEL · CALCULATION NOTE

VatReel sizing and first-principles checks

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Rerun on the constructable design (VTR-DDR-002); torque tube, guard case, wider shield, extension tube 1.55 m; cost USD 1,437
0.1	2026-10-03	Amish Chadha	First TRL 3 sizing note against every requirement (depth, span, drive, brake, lift, structure, masses, splash, cost)

VatReel sizing and first-principles checks

On paper VatReel does what it is meant to do: every carrier bar comes up above the rim at every working depth, so hides and yarn are loaded and unloaded without a hand in the liquor; the reel turns with about 90 N at the crank with the whole 25 kg wet load on one bar; the lowest bar can be set anywhere from 300 to 1,000 mm below the rim; and the self-locking lift screw holds the reel at any height and lifts every bar clear of the rim. Ten of the twelve requirements are met on paper. R8 (portability) is not met as written: two modules weigh 27.1 and 29.6 kg, so they are two-person carries, and first setup is estimated at about 29 minutes against 15. R6 (corrosion) can only be shown by a coupon test and is at risk for the stainless chain in chloride pickling liquor. The parts cost about USD 1,437 against the USD 1,500 value-engineering target, USD 63 under it.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`). The script reads the geometry from `PARAMS` and `geometry()` in `cad/src/model.py`, takes masses from the model solids and reads prices from `bom/bom.csv`, so the model, the drawing VTR-DWG-001, the BOM and this note agree. All values are first-principles estimates; nothing here is measured.

Safety: VatReel is moving machinery used beside open pits of corrosive and, in dyeing, hot liquor (up to 98 °C (208 °F)). The checks below are paper checks of strength and reach; they do not replace a risk assessment of the site, the pit edge or the process chemistry.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Wet load	25 kg of hides or yarn with the liquor they carry	R2
Worst case for torque	Whole wet load on one carrier bar level with the axle; buoyancy and drag ignored	Conservative
Proof factor	1.5 x the rated load; stresses held to yield / 1.5 at proof	R5
Pit	1.0 to 2.5 m wide, at least 1.9 m long, rim level with the floor, liquor 150 mm below the rim	VTR-PRB-001; site survey to confirm
Materials	316L tube and sheet, yield 170 MPa; 316 bar and 304 sections, yield 205 MPa; E = 193 GPa	Minimum 0.2 % proof stresses
Chain	ISO 606 08B-1, minimum tensile strength 18.0 kN	Standard
Chain stage efficiency	0.95 per stage including its plain bearings	Typical
Axle bearings	Glass-filled PTFE on stainless, friction 0.10 wet	Typical
Lift screw	Tr24 x 5 single start, bronze nut, thread friction 0.15 (wet, dirty), thrust ball bearing at the top	Conservative
Operator	Eye 1,600 mm high, 1,150 mm behind the near pit edge, at the crank	1.75 m person
Cranking speed	30 rpm	Steady hand cranking
Prices (2026)	304 stainless USD 4.50/kg, 316 and 316L USD 6.50/kg; bought parts at regional retail estimates	Indicative, not quotes

■ 2. Depth and positions (R1, R4, R11)

The arms swing on the pivot line 200 mm above the rim; the arm angle sets the depth. Every working position keeps the top carrier bar above the rim, which is where loads are hung and taken off.

Table 2. Arm positions (heights from the rim; + is up; mm).

Position	Arm angle below level	Axle height	Lowest bar	Top bar	Reel extends along the pit	Screw length
Lifted	-33.7 deg	+699	+149	+1,249	159 to 1,339	1,020
Shallowest working	-3.2 deg	+250	-300	+800	309 to 1,489	894
Mid depth (modelled)	+20.0 deg	-108	-658	+442	256 to 1,436	795
Deepest working	+46.2 deg	-450	-1,000	+100	33 to 1,213	704

R4 is met: the lowest bar can be set anywhere from 300 to 1,000 mm below the rim. R1 is met on paper: the top bar is 100 to 800 mm above the rim at every working depth, and lifted, every bar is at least 149 mm above it. The model's fit check confirms that nothing collides with the frame or the pit walls at the lifted, shallowest and deepest positions (`python cad/src/model.py --check`).

■ 3. Spans (R3)

The pivot tube alone reaches pits 1,000 to 1,220 mm wide; with the extension tube pinned inside it, 1,220 to 2,520 mm. R3 is met. The reel needs a pit at least 1,890 mm long (the pivot 350 mm in from the end wall, plus the reel swing and 50 mm clear); this is added to the assumptions in VTR-REQ-001. The drum is 600 mm long and the far end of the arm frame is 955 mm in from the near edge, so a 1.0 m pit leaves 45 mm clear.

■ 4. Drive (R2)

Table 3. Drive at the worst case.

Item	Value
Load torque on the reel (25 kg at 550 mm)	134.9 N m
Axle bearing friction	1.3 N m
Ratio, chain 1 (15 to 30 teeth) x chain 2 (17 to 51 teeth)	6.0
Torque on the jackshaft	47.8 N m
Torque on the crank shaft	25.1 N m
Hand force at 280 mm crank radius	90 N
Chain 1 tension; factor on breaking load at proof	823 N; 15
Chain 2 tension; factor on breaking load at proof	1,383 N; 9
At 30 rpm on the crank: reel speed; carrier bar speed; time per turn	5.0 rpm; 0.29 m/s; 12 s

R2 is met: 90 N against 100 N in the worst case. With loads spread round the reel the torque is a fraction of this.

■ 5. Holding brake (R5)

The ratchet on the crank shaft holds 1.5 x the worst load torque with no credit for chain friction: 33.7 N m on the crank shaft, a tooth force of 602 N, 12.5 MPa bearing on the 8 mm tooth, 5.3 MPa shear in the 12 mm pawl pin, and 6.4 MPa (crank shaft) and 5.4 MPa (jackshaft) in torsion. R5 is met on paper with large margins; the CalRig proof-load test is TRL 4 work.

■ 6. Lift screw (R11)

The Tr24 x 5 thread has a lead angle of 4.23 deg against a friction angle of 8.83 deg, so it is self-locking: the reel cannot run down on its own. Its stroke is 316 mm, 63 turns from lifted to deepest.

Table 4. Lift screw at 1.5 x the load moment.

Position	Screw length (mm)	Lever arm (mm)	Moment (N m)	Screw force (N)	Torque (N m)	Handwheel rim force (N)
Lifted	1,020	214	704	3,298	8.5	68
Shallowest	894	250	845	3,386	8.7	70
Mid depth	795	233	795	3,410	8.8	70
Deepest	704	152	586	3,847	9.9	79

The screw is in compression; its Euler load at the longest length is 10.5 kN against 3.85 kN (factor 2.7), and the root stress is 14.3 MPa. R11 is met on paper.

■ 7. Structure at the proof load

Table 5. Stresses at 1.5 x, against yield / 1.5.

Part	Load case	Stress (MPa)	Yield (MPa)	Factor on yield
Carrier bar	Whole load spread along one bar, simply supported	19	170	9.2
Spoke	Half that load at the tip, no help from the hexagon ties	106	170	1.6
Reel axle	Reel and load at mid span plus torque	43	170	4.0
Arm	Half the reel and load at its end, arm level	40	170	4.3
Torque tube	Far arm's moment carried to the lever	29	170	5.8
Jackshaft	Overhang carrying the near bush and the pivot tube plug (242 N m)	39	205	5.3
Pivot tube	2.5 m pit, far bush load between plug and clamp (299 N m)	25	205	8.2
Lift screw bracket	3.85 kN at 255 mm	118	205	1.7

Every part is within yield / 1.5 at the proof load. The spoke and the bracket have the smallest margins; the spoke figure ignores the ties, which share the load.

■ 8. Support

The centre of the reel stays well inside the feet of both stands (451 mm at the lifted position, 577 mm at the deepest). With the reel loaded, the near stand carries 316 to 550 N and the far stand 471 to 237 N across pits 1.0 to 2.5 m wide; both are small loads on a sound rim.

■ 9. Masses and setup (R8)

Table 6. Masses of the parts carried to the pit (kg).

Module	Mass	Carry
Near stand base frame with pads and hooks	19.5	One person
Bearing tower, bare	23.2	One person
Shafts, sprockets, chain 1, crank and ratchet (fitted in place)	16.2	Several pieces
Tower braces	4.0	One person
Splash shield	12.1	One person
Lift screw bracket	3.9	One person
Lift screw with trunnions and handwheel	8.6	One person
Pivot tube with the arm frame on it	27.1	Two people, 13.5 each
Extension tube	6.3	One person
Far stand	12.4	One person
Chain 2 with its guard	8.2	One person
Reel with carrier bars and axle sprocket	29.6	Two people, 14.8 each
Total	171.2	

R8 is not met as written: two modules exceed 25 kg, though no person carries more than 23.2 kg. First setup is estimated at 29 minutes over ten steps (`setup_time()` in the script) against 15; moving between pits of the same width is quicker. Both are recorded as decided limits in VTR-DEC-001.

■ 10. Splash (R9)

All 156 straight lines checked from the top half of the reel, at four depths and three positions along it, to the operator's eye pass through the splash shield (crossings 89 to 1,427 mm above the rim, inside the shield's 60 to 1,450 mm) or into the near pit wall. R9 is met on paper; droplets do not travel in straight lines, so the indicator paper trial remains the test (TRL 4).

■ 11. Heat and corrosion (R6, R7)

Table 7. Material limits.

Material	Where	Limit	Against 98 °C liquor
316L stainless	Reel, arms, wetted parts	Well above 98 °C	Met
Glass-filled PTFE	Axle bushes in the liquor	About 260 °C	Met
UHMW-PE	Bushes above the liquor	About 80 °C continuous	Met if splash only; temperature at the bushes to confirm
Polypropylene	Shield sheet	About 100 °C	Met (splash only)
EPDM	Pads	About 120 °C	Met

R7 is met on paper. R6 cannot be shown by calculation: 316L resists lime and sulphide liquors and chrome liquors well, but chloride pickling liquor can pit stainless steel, and the 304 chain 2 is the weakest part. It is marked at risk until the coupon immersion test.

■ 12. Cost (R10)

The priced BOM totals USD 1,437. Value-engineering target: USD 1,500. Estimated cost of the constructable design: USD 1,437 (USD 63 under the target).

■ 13. Results against the requirements

Table 8. Requirements on paper.

ID	Requirement	Result on paper	Status
R1	No hand in the liquor	Top bar 100 to 800 mm above the rim at every working depth; loaded with hooks	Met on paper
R2	Crank force 100 N or less, 25 kg	90 N worst case	Met
R3	Pits 1.0 to 2.5 m wide	1.0 to 2.52 m; pit at least 1.89 m long	Met
R4	Lowest carrier 0.3 to 1.0 m deep	0.30 to 1.00 m, any depth between	Met
R5	Brake holds 1.5 x	Ratchet stresses under 13 MPa; self-locking lift screw	Met on paper
R6	Corrosion, 500 h in three liquors	Material choice only	At risk (chain 2 in pickle)
R7	98 °C baths	All materials rated	Met on paper
R8	Modules 25 kg or less; setup 15 min	Two modules 27.1 and 29.6 kg (two-person carries); about 29 min	Not met
R9	No droplets reach the operator	All sight lines blocked	Met on paper
R10	Cost against the USD 1,500 target	USD 1,437	USD 63 under the target
R11	Self-locking lift, 100 N or less	79 N at 1.5 x; self-locking	Met on paper
R12	Moving parts guarded	Chains, sprockets and ratchet enclosed	Met on paper; openings to check